Project Proposal

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Problem

I work on the Vehicle Routing Problem (VRP). A single depot, a fleet of vans, and a set of customer nodes must be visited. Each customer requests a specific amount of some good (e.g., a supermarket ordering loaves of bread from a bakery), and each van has a limited carrying capacity. The goal is to find one route per van, starting and ending at the depot, visiting every customer exactly once without exceeding any van's capacity, while minimizing total distance traveled. The simplest special case is the TSP: a single vehicle, no capacity, no demand.

Input: The input consists of a set of nodes comprising the depot and customer locations, where each customer j has a demand q_j. The fleet includes K available vans, each with a maximum load capacity Q_k, alongside a distance matrix d_{ij} defining the travel cost between every pair of nodes.

Output: Minimizing total distance traveled across all vans.

Formulation: Let $x_{i,j,k} \in \{0, 1\}$ indicate van k travels directly from node i to node j.

- **Objective:** $\min \sum_k \sum_{i,j} d_{ij} \cdot x_{i,j,k}$
- **Visit each node once:** $\sum_k \sum_i x_{i,j,k} = 1 \quad \forall j \neq depot$
- **Flow conservation:** $\sum_j x_{i,j,k} = \sum_j x_{j,i,k} \quad \forall i \neq depot, \ \forall k$
- **Depot degree:** $\sum_j x_{depot,j,k} \leq 1, \quad \sum_i x_{i,depot,k} = \sum_j x_{depot,j,k} \quad \forall k$
- **Van capacity:** $\sum_i \sum_{j \neq depot} q_j \cdot x_{i,j,k} \leq Q_k \quad \forall k$

Implementation

Heuristic baseline: The Clarke–Wright Savings algorithm (Clarke & Wright, 1964) is the baseline.It initializes one van per node, then repeatedly merges the pair of routes with the largest savings $s_{ij} = d_{i,depot} + d_{depot,j} - d_{ij}$, subject to capacity, until no feasible merge remains.

RL approach: Two learning methods will be built on top of the same state representation. First, a self-attention model will turn each node's features (coordinates, demand) and the depot into embeddings, plus one embedding summarizing the whole instance – computed once per instance, before any node is chosen. Then, at each step, the model will compare the current situation (instance summary, current node, remaining van capacity) against every node's embedding to get one score per node. Nodes that are already visited or that would exceed the van's remaining capacity will be blocked out (score set to $-\infty$) so they can never be chosen, which lets the same model handle any number of remaining nodes.

Two versions of this model will be trained and compared:

- **Value-based:** each node's score will be treated as a Q-value and trained with DQN.
- **Policy-based:** each node's score will be turned into a probability with softmax and trained with A2C.

Reward will be the negative distance of each edge taken, which adds up to the negative total route length by the end. Since both versions use the same self-attention model underneath, this setup will show whether the value-based or policy-based version works better, in addition to comparing both against the Clarke–Wright heuristic baseline above.